

Akhilesh Babu Tumati

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PROFESSIONAL EXPERIENCE

Software Engineer, Tata Consultancy Services

Aug 2023 - Aug 2024 — Hyderabad, India

- Automated Jenkins CI/CD pipelines for UPS client, accelerating deployment speed by 40% and decreasing critical deployment errors by 30% through comprehensive automated testing and rigorous validation protocols.
- Maintained Couchbase NoSQL database infrastructure, achieving a 25% improvement in data retrieval efficiency, which bolstered overall application performance by 15% for the client.
- Developed a robust data quality assurance framework that decreased data discrepancies by 40%, critically enhancing the accuracy and reliability of analytics outputs for strategic planning and operational decision-making.
- Implemented comprehensive system monitoring using DynaTrace and Grafana, resulting in a 20% reduction in system downtime and a 10% improvement in application response time through proactive issue identification and resolution.

Software Developer, Honeywell

Jan 2023 - Jul 2023 — Hyderabad, India

- Spearheaded the creation of automated testing scripts utilizing Katalon Studio, resulting in a 30% reduction in testing cycles across enterprise applications and 25% expansion of test coverage.
- Enhanced front-end applications using JavaScript and Angular frameworks, thereby delivering 20% faster page load times and boosting user satisfaction metrics by 15% across platforms.
- Implemented Coverity and Blackduck security protocols, fortifying application security standards and elevating code quality metrics by 40%, thereby mitigating potential vulnerabilities. This also allowed for quicker resolution of defects.
- Enhanced application performance through Dynatrace monitoring, achieving a 35% surge in system efficiency and a 20% reduction in critical system downtime, reducing customer-reported issues by 15%.

EDUCATION

Masters in Computer Science

Aug 2024 - Present — Virginia, United States

Virginia Polytechnic Institute and State University: GPA: 4.0/4.0

Bachelors in Computer Science and Engineering

Jul 2019 - Aug 2023 — Andhra Pradesh, India

Vellore Institute of Technology: GPA: 3.5/4.0

SKILLS

Programming Languages: Java, JavaScript, Python, SQL, Maven, XML, Hibernate, JUnit.

Software Development: TypeScript, React, Next.js, Node.js, Spring Boot, Sanity CMS, Katalon, Selenium, Jira, DynaTrace, Grafana, Coverity, Jenkins CI/CD, Bamboo, BitBucket, Git, Confluence, AWS (EC2, VPC, EKS, Lambda).

Databases: MySQL, CouchBase, MongoDB, Oracle SQL Developer.

Data Science: NumPy, Pandas, Matplotlib, TensorFlow, scikit-learn, PyTorch, PowerBI, Keras.

PROJECTS

Technoverse Blog,

Nov 2024 - Dec 2024

- Developed a high-performance blog platform using TypeScript and Next.js on Vercel, integrating Sanity CMS to optimize user experience and content management efficiency.

Mahabharat Character Detection,

Aug 2022 - Dec 2022

- Designed a comprehensive character detection pipeline employing pytorch, TensorFlow, CNN models, ViT, and Yolov5 achieving a 30% boost in accuracy and reducing false positives by 20% in complex video environments.

Crypto Currency Value Prediction,

Sep 2021 - Nov 2021

- Engineered advanced time series prediction models using Python, achieving 85% accuracy in cryptocurrency value forecasting using ARIMA and SARIMA algorithms, enabling real-time analysis through a Streamlit dashboard.

Creating Private Cloud Storage:

May 2021 - Jul 2021

- Architected scalable private cloud infrastructure using Raspberry Pi, NextCloud, Python, and HardDiskDrive for storage. Scheduled secure storage protocols and streamlined data transfer mechanisms for enhanced performance.

PUBLICATIONS

[1] Tumati, A. B., Gangaraju, R., Mannepalli, B. R., Sumathi, D., & Alluri, B. K. (2023, January). Face Invariant Classification and Detection of Mythology Characters Using Custom Dataset (ClaDeMuC-CD). In *2023 Third International Conference on Advances in Electrical, Computing, Communication and Sustainable Technologies (ICAECT)* (pp. 1-8). IEEE. 15 May 2023

[2] Natarajan, K., Gangaraju, R., & Tumati, A. B. (2022). Hybrid ML and DL models for flood level prediction. *International Journal of Health Sciences*. 08 Dec 2022